

The Basics of Piping

Learning how to pipe using different tips

Sofia Hirao | COMS W4170 | Spring 2026

Domain: Baking

Topic: Learning how to pipe using different tips

Expert: Sofia Hirao (Baker who has taken many baking classes)

User: Jolie, 19, student, wants to learn how to make cakes for her friends birthdays



Domain: Baking

Group Member: Sofia Hirao

Topic: Basics of piping techniques

This topic meets the requirements because it is a focused, visual skill that can be taught in 10 minutes through demonstration and practice.

Expert: Sofia

- Jolie: Has baked before and used a piping tip but has struggled in the past.
- Sam: does not really bake and is not familiar with these terms and usages.

Media: What media would you use?

- 1) Step-by-step video demonstrations of each piping tip in action.
- 2) Pictures showing the final designs for each tip.
- 3) short animated GIFs showing the pressure and motion for piping.

What is one positive piece of feedback your TA gave you on this idea?

- This is something teachable in 10 minutes and it is something someone would want to know

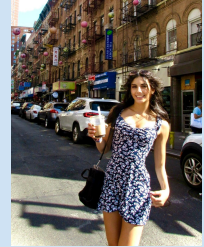
What is one negative or cautionary piece of feedback your TA gave you on this idea?

- This is something that requires following along and feedback to make sure they are doing it correctly so keeping that in mind when thinking about this topic.

User:

- Jolie Abadir is a current sophomore at Barnard College, majoring in Computer Science.
- Jolie is interested in learning how to use different piping tips because she wants to bake and decorate cakes for her friends' birthdays. For example, she recently wanted to make cupcakes for a friend's party but couldn't create decorative frosting patterns, so learning proper piping techniques would let her create visually appealing desserts.

PICTURE



Quiz:

- Match the piping tip to its frosting design
- Identify how much pressure is required for each pattern
- Simulate piping a shape using a drag and drop or hover over activity to practice technique

Who will you run your prototype on?

- I will run the prototype with another student in class who has minimal baking experience and is interested in learning cake decorating.

Baking - Low-fidelity prototype

Idea: Basics of piping techniques how different tips create different designs

- **Person**

- Jolie a student who wants to bake for friends

- **Goal**

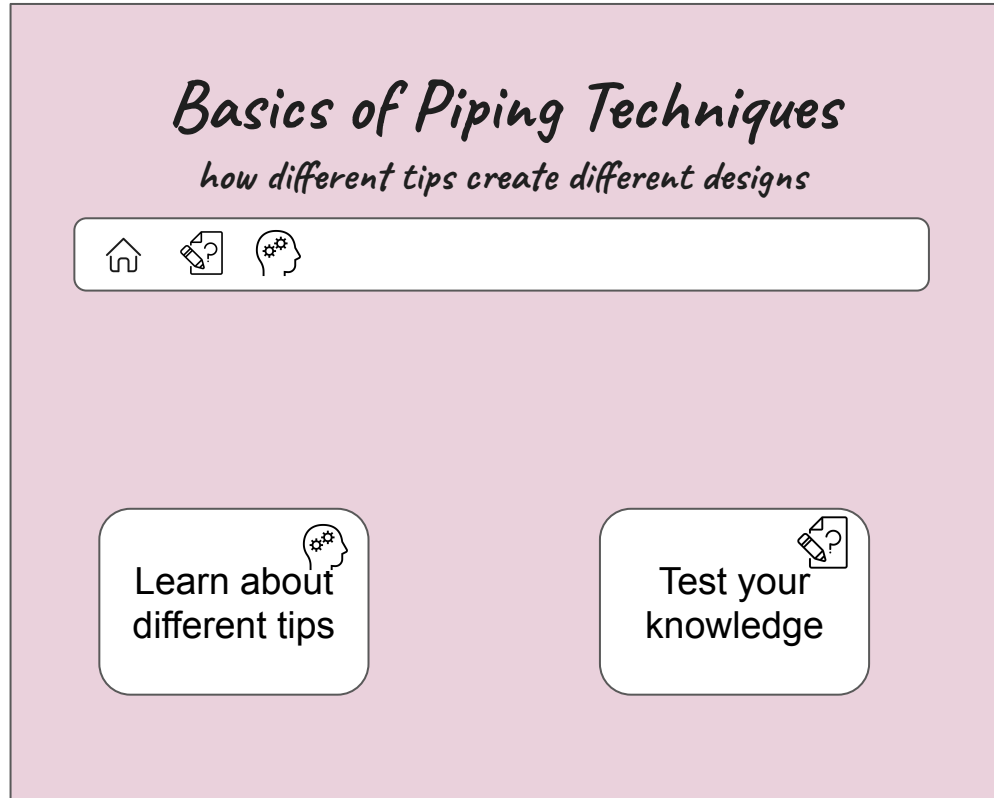
- Teach them how to use different tips for Piping

- **Subgoals**

- Learn hallmarks of different tips
- Learn how to pipe using the different tips
- Pass a final quiz

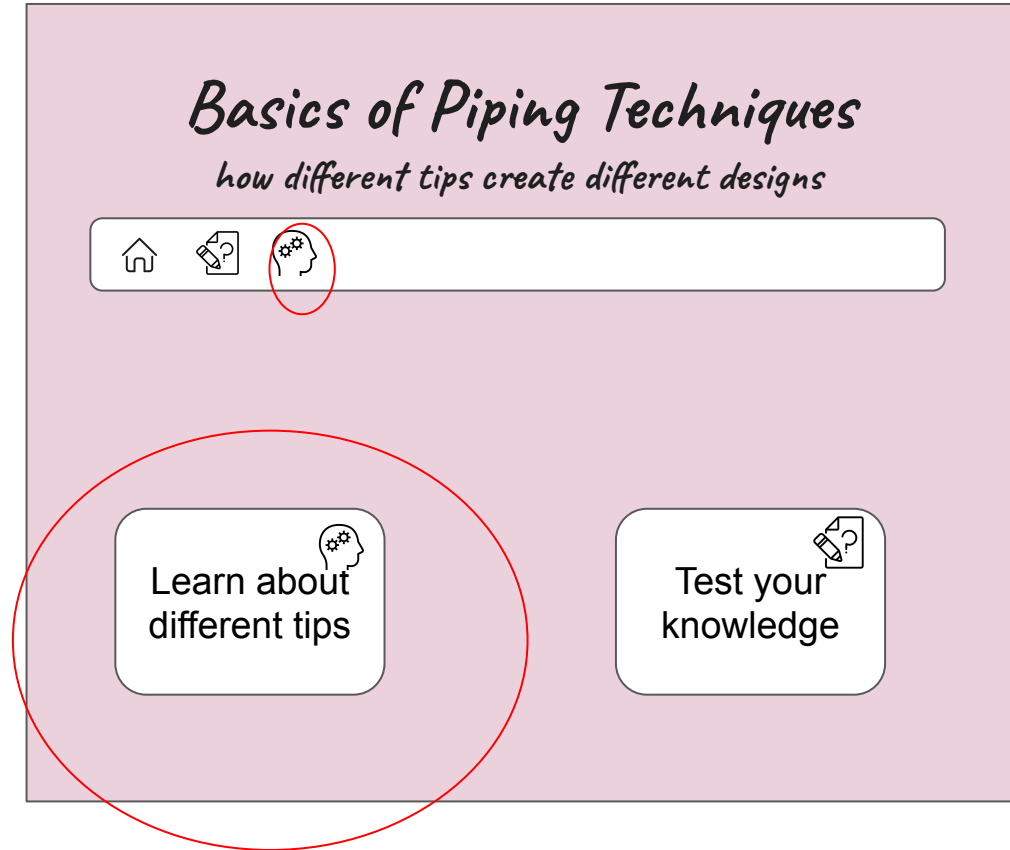
Subgoal 1: Learn hallmarks of different tips

1) Home Screen



Subgoal 1: Learn hallmarks of different tips

2) Select Learn about Piping Tips



Subgoal 1: Learn hallmarks of different tips

3) Learn about different Piping Tips

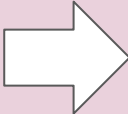
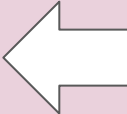
Basics of Piping Techniques



The Round Tip

The Hallmarks: Explain that this tip has a plain, circular opening with no teeth or ridges. This is for "fine motor" work like writing names or creating simple borders. The Round Tip





Subgoal 2: Learn how to pipe using the different tips

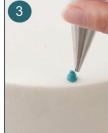
Step by Step how to pipe

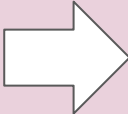
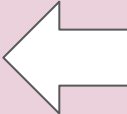
Basics of Piping Techniques



The Round Tip Piping steps

- 1 Using a decorating bag fitted with a round tip and medium consistency icing, hold the bag straight up, with the tip just slightly above the surface.
- 2 Squeeze the bag to form a dot. As the icing begins to build up, raise the tip with it while keeping the tip point slightly buried in the icing.
- 3 Stop squeezing and pull the tip up slightly, but not away from the icing. With a circular motion, use the end of the tip to scrape the point on top of the dot.
- 4 To create a pull-out dot, form the dot, then continue squeezing and pull tip up and away. Decrease pressure as you reach the desired length.





Subgoal 1: Learn hallmarks of different tips

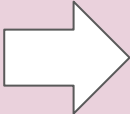
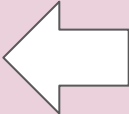
3) Learn about different Piping Tips

Basics of Piping Techniques



The Star Tip

The Hallmarks: Point out the "teeth" or "v-shape" cuts in the metal. The number of teeth determines how many ridges the frosting has. This is the go-to for "fancy" looking cupcakes.



Subgoal 2: Learn how to pipe using the different tips

Step by Step how to pipe

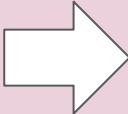
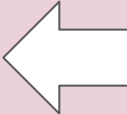
Basics of Piping Techniques



The Star Tip

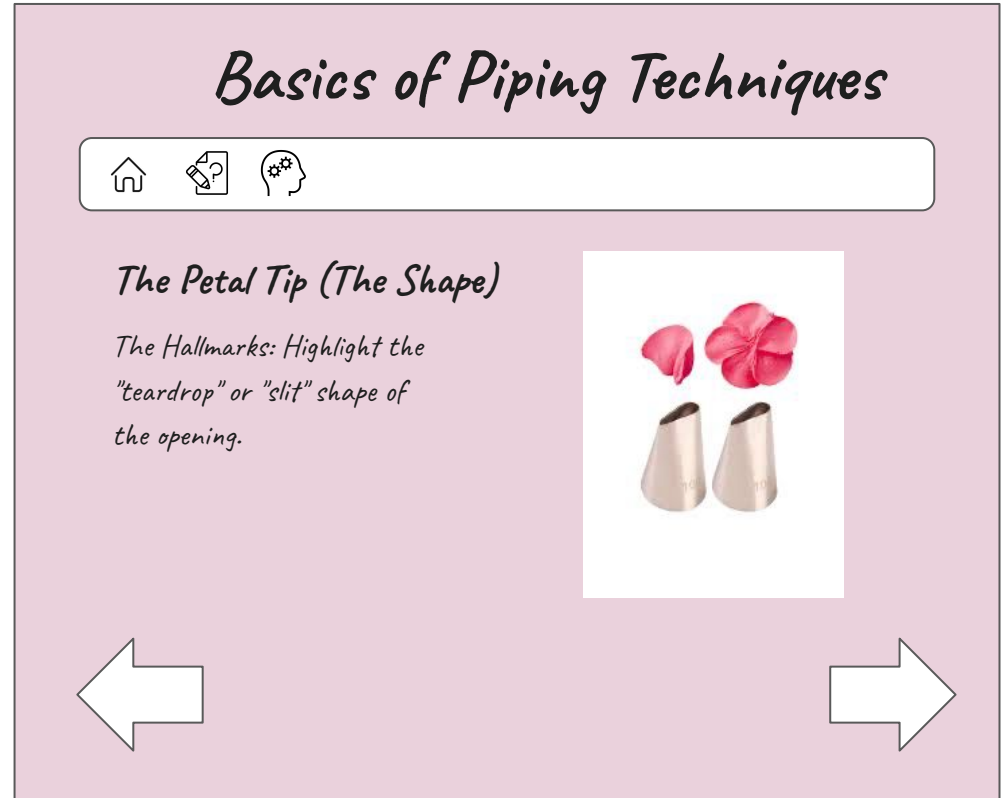
- 1 Using a decorating bag fitted with a star tip, hold the bag straight up, with the tip just slightly above the surface.
- 2 Squeeze the bag to form a star. Use light pressure to form a small star; add more pressure to make a bigger star.
- 3 Stop squeezing before pulling tip away.
- 4 To create a pull-out star, form the star, then continue squeezing and pull tip up and away. Decrease pressure as you reach the desired length.





Subgoal 1: Learn hallmarks of different tips

3) Learn about different Piping Tips



Subgoal 2: Learn how to pipe using the different tips

Step by Step how to pipe

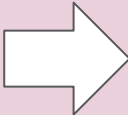
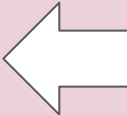
Basics of Piping Techniques



The Petal Tip (The Shape)

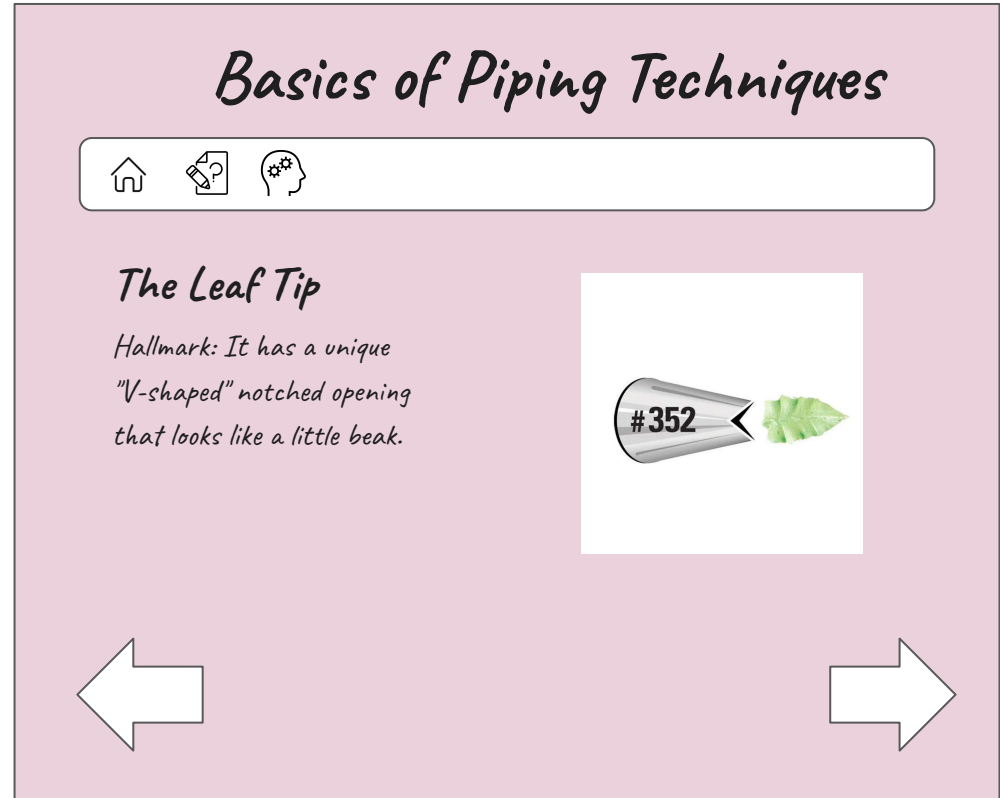
- 1 Using a decorating bag fitted with a drop flower tip, hold the bag straight up, but bend your wrist inward.
- 2 With the tip just touching the surface, squeeze out icing.
- 3 As you squeeze, turn your wrist outward.
- 4 Stop squeezing before pulling tip away. Pipe a center if desired using tip 2, 8 or 18.
- 5 Hint: Using different tips will create different flower effects. Tip 10 is perfect for making simple daisies, whereas tip 2F creates petals with interesting dimensions.





Subgoal 1: Learn hallmarks of different tips

3) Learn about different Piping
Tips - Tip Picture & Shape it
makes just frosting



Subgoal 2: Learn how to pipe using the different tips

5) Step by Step how to pipe

Basics of Piping Techniques

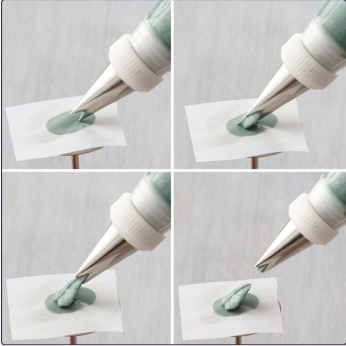


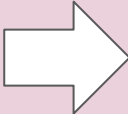
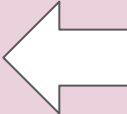
The Leaf Tip

- 1 Leaves add extra details to flower arrangements, and they don't always have to be green! Leaves can be piped in different colors and lengths, creating different effects.

Icing Consistency: Medium

- 2 On a flower nail or a flat surface, hold a decorating bag fitted with tip 352 at a slight upward angle from the surface. The two points of the tip must be vertical (so that it looks like an open beak).
- 3 Squeeze hard to form the base of the leaf.
- 4 As you squeeze, slowly pull the tip slightly upward (this will give the leaf a little lift).
- 5 Relax pressure as you pull away to form a leaf point. Stop squeezing before pulling tip away.

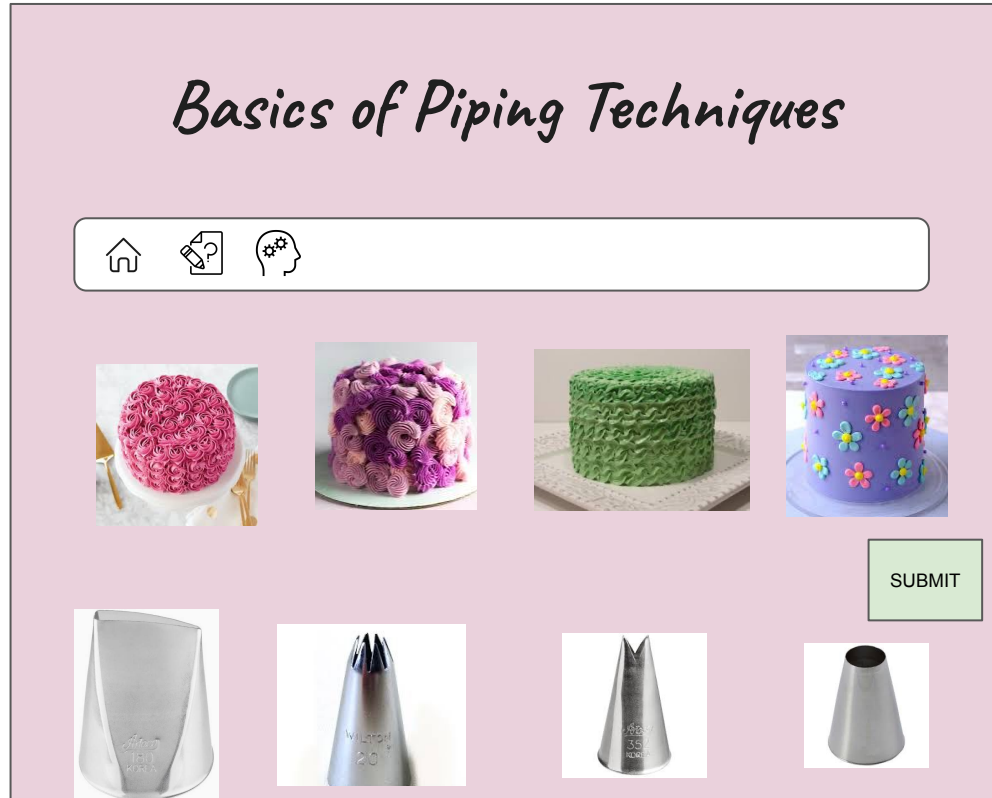




Subgoal 3: Pass quiz:

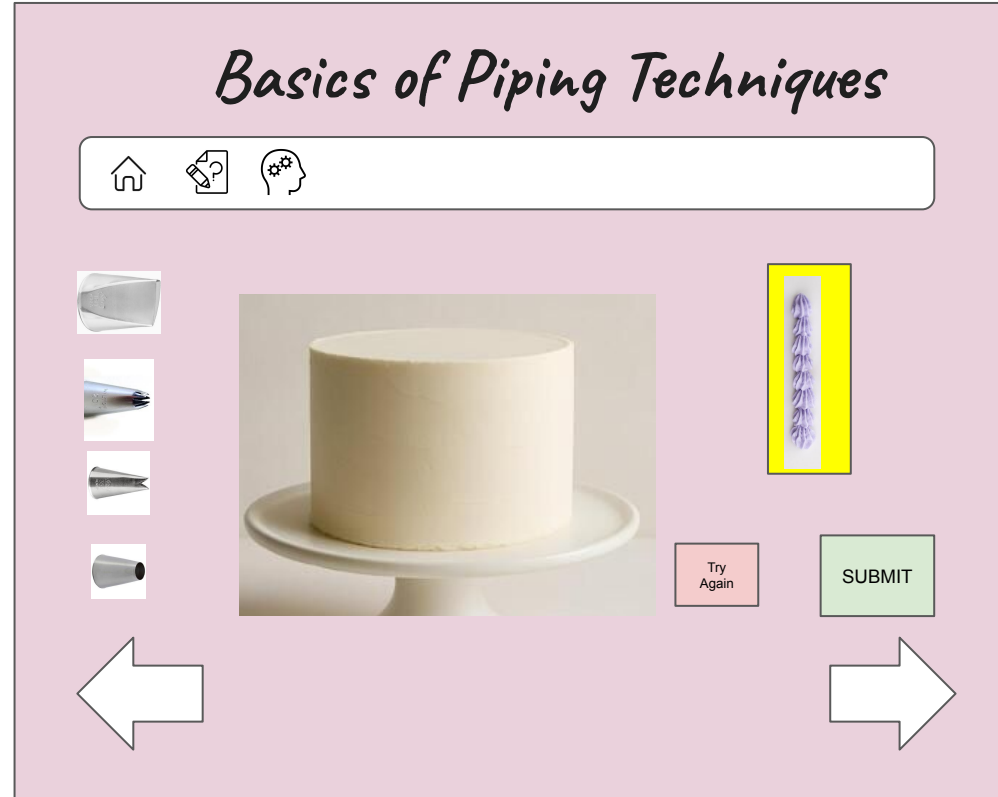
Identify the Tip

Where the user drag the correct tip to the cake then presses submit when they are finished.



Subgoal 3: Pass quiz

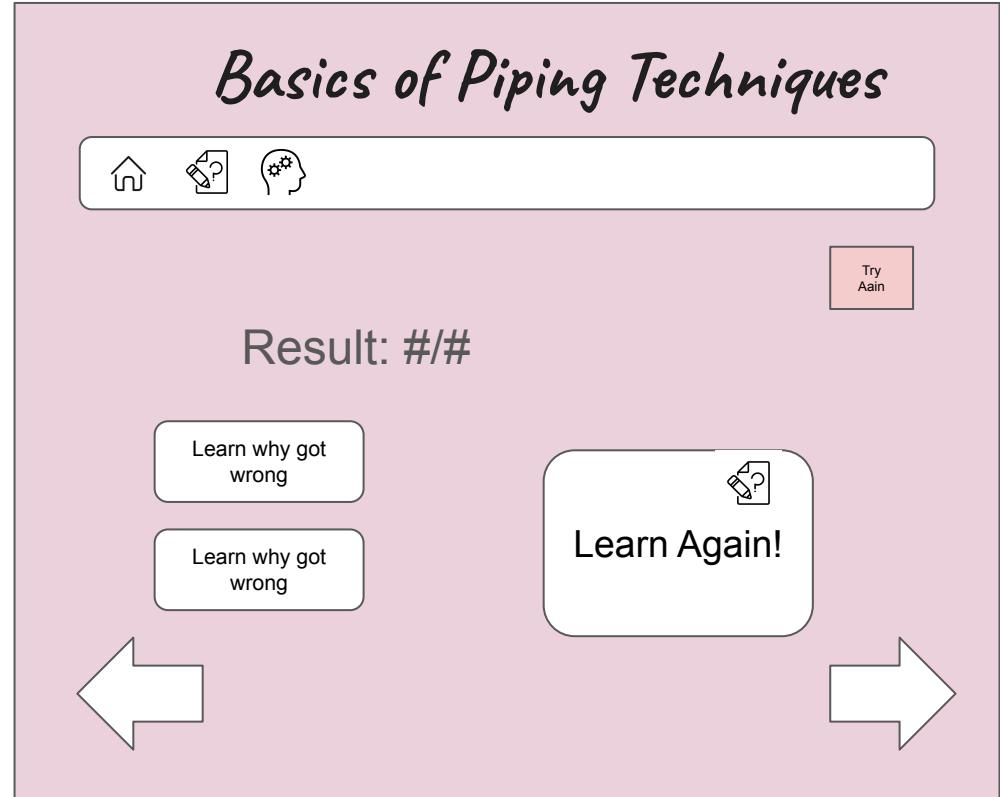
The user is given an undecorated cake and they will be shown a line of frosting piped from a specific tip and the user will need to pick the tip correctly then do the motions of piping on the cake. There will be a try again button and a submit button.



Subgoal 3: Pass quiz

The result of the Quiz with the tips gotten wrong from the drag and drop section and the piping section.

For everything the user got wrong there will be a explain why button.



Insights

- **Positive:** User found the visual hallmarks of the Star tip very easy to remember
- **Negative:** The user was confused about where to click on the piping simulation. Without a clear Start indicator on the cake she did not know if she should pipe or not
- **Negative:** The feedback on the Incorrect slides was too brief. The user wanted to see a side-by-side comparison of the design they should have made versus the one they actually picked to understand the specific error.
- **Negative:** Simulating the physical pressure of a piping bag was difficult for the user to grasp through a mouse click. Jolie felt that clicking a button didn't accurately represent the "squeeze and release" required for piping